MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL, HEADER, RIGHT ANGLE (.100 SPACING) WITH POLARIZING FEATURE SOLDERLESS WRAP POST OR PWB TERMINATION

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the header connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-83503(USAF)

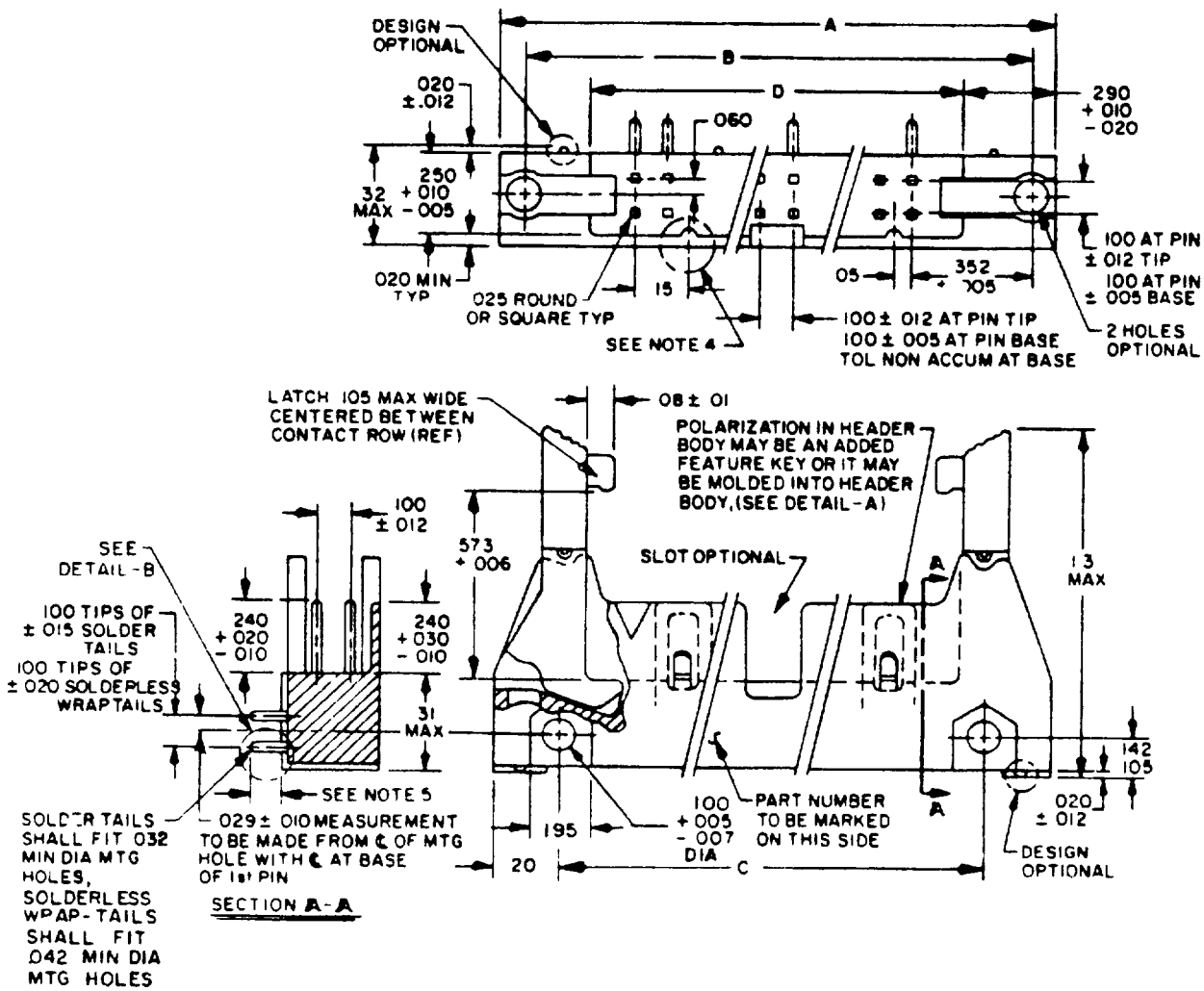
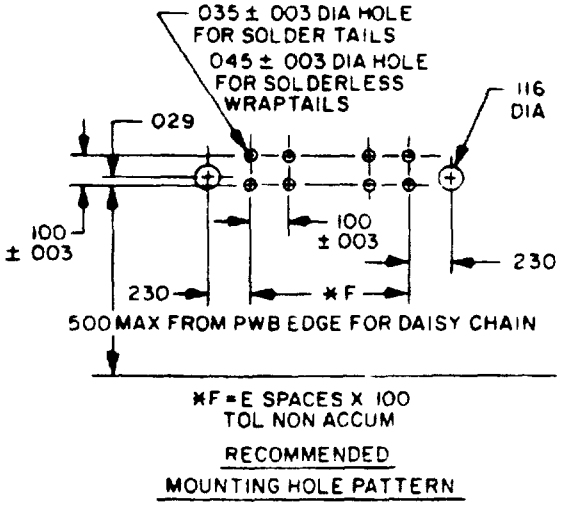
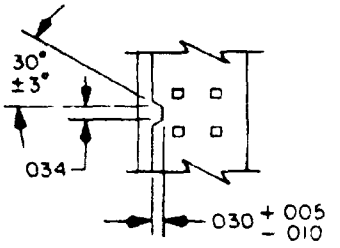


FIGURE 1 Dimensions and configuration

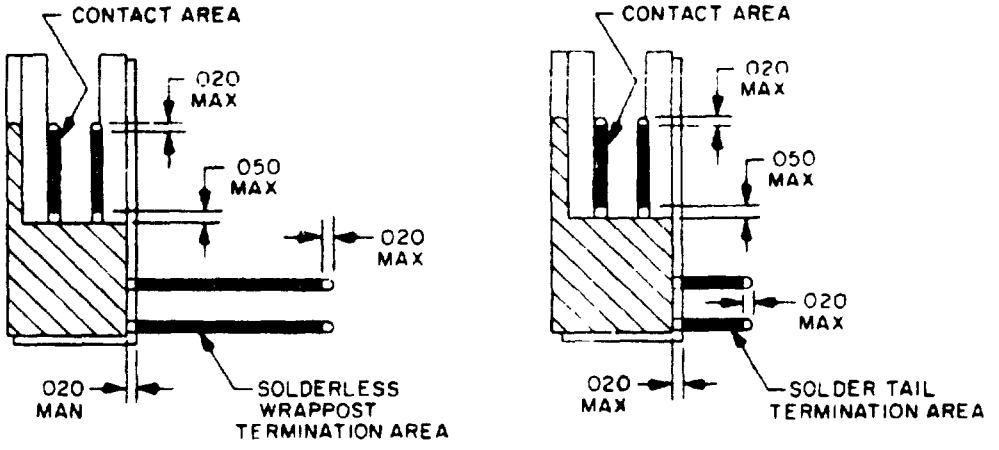
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RECOMMENDED
MOUNTING HOLE PATTERN



POLARIZATION DETAIL
DETAIL-A



FINISH LOCALIZATION AREA
DETAIL-B

FIGURE 1. Dimensions and configuration - Continued

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Inches	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.029	.74	.10	2.5	.230	5.84
.005	.13	.030	.76	.100	2.54	.240	6.10
.006	.15	.034	.86	.105	2.67	.250	6.35
007	.18	.035	.89	.116	2.95	.290	7.37
010	.25	.045	1.14	.125	3 18	.31	7.9
01	.3	.05	1.3	.142	3.61	.32	8.1
.012	.30	.060	1.52	.15	3.8	.352	8.94
.015	.38	.062	1.57	.152	3.86	.500	12.70
020	.51	08	2 0	.178	4.52	.573	14.55
025	.64	.089	2.26	.195	4.95	.610	15.49
				.20	5.1	1.3	33.0

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ± 01 (.3 mm) for two place decimals
4. The polarizing feature on this end is non-existent on the 10 position connector and 14 position connector.
5. For 062 PWB pin length shall be .089 minimum and .116 maximum, for 125 PWB, pin length shall be .152 minimum and .178 maximum, and for solderless wrappost, pin length shall be .610 ±.015.

FIGURE 1 Dimensions and configurations - Continued.

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TABLE I. Dash numbers and characteristics

Dash number	No of pins	Dimensions 1/ A	B	C	D +.030 -.000	E	F	Termination, right angle contact type
-01	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	062 PWB
-02	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	125 PWB
-03	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	Wrappost
-04	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.062 PWB
-05	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.125 PWB
-06	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	Wrappost
-07	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.062 PWB
-08	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.125 PWB
-09	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	Wrappost
-10	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	.062 PWB
-11	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	125 PWB
-12	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	Wrappost
-13	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	062 PWB
-14	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	.125 PWB
-15	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	Wrappost
-16	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	062 PWB
-17	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	.125 PWB
-18	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	Wrappost
-19	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	.062 PWB
-20	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	125 PWB
-21	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	Wrappost
-22	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	062 PWB
-23	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	125 PWB
-24	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	Wrappost
-25	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	.062 PWB
-26	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	125 PWB
27	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	Wrappost
-28	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	062 PWB
-29	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	.125 PWB
-30	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	Wrappost

1/ Metric equivalents are in parentheses.

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REQUIREMENTS:

Design and construction

Dimensions and configuration See figure 1 and table I.

Temperature range -55°C to +120°C.

Voltage rating. 300 V rms.

Current rating: 1 ampere.

Materials.

Contacts. Phosphor bronze or brass per CDA260, spring temper, or copper alloy CA-725 per ASTM-B-122.

Contact plating
Contact area: Gold
Solderless wrappost terminal area: Per MIL-STD-1130.
Solder tail termination area. Gold, or tin-lead.

Housing Glass filled polyester per MIL-M-24519, type GPT-15F, GPT-20F or GPT-30F

Accessories Parts may be constructed of polyester per MIL-M-24519 or other materials specified in MIL-C-83503, provided these parts do not come in contact with current carrying components.

Polarization key: Shall be permanently attached to header (added feature key shall be bonded to header).

Mating connector See MIL-C-83503/7

Solderless wrappost contacts Geometry parallelism and tip configuration shall be in accordance with MIL-STD-1130.

Solderless wrappost contact wire accommodation 26 or 28 AWG solid.

Dielectric withstanding voltage (at sea level): 500 V rms, minimum.

(at altitude) 200 V rms, minimum.

Cable retention Not applicable

Contact retention Not applicable.

Connector-cable flexing Not applicable.

Contact resistance 50 milliohms, maximum

Latch retention The latch shall retain the mating connector when the cable retention force is applied to the mating connector cable The latch shall also function as an ejector to eject mating connector.

Marking

Part number M83503/20-(dash number from table I).

Contact identification Molded triangle "A" or notch to show number 1 contact, or raised numbers to show position number "1".

NOTE Revision letters are not used to denote changes due to the extensiveness of the changes

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Air Force - 11, 17, 99
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Agent
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Preparing activity
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